



UNIVERSITY^{AT}ALBANY
STATE UNIVERSITY OF NEW YORK

CSI 436/536 (Spring 2026)

Machine Learning

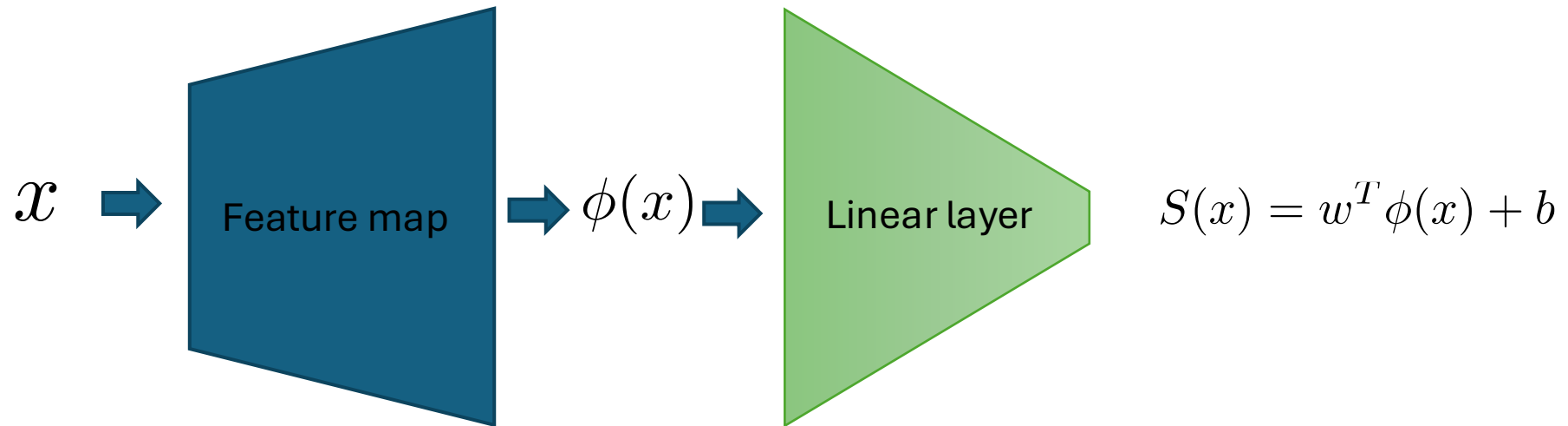
Lecture 17: Clustering

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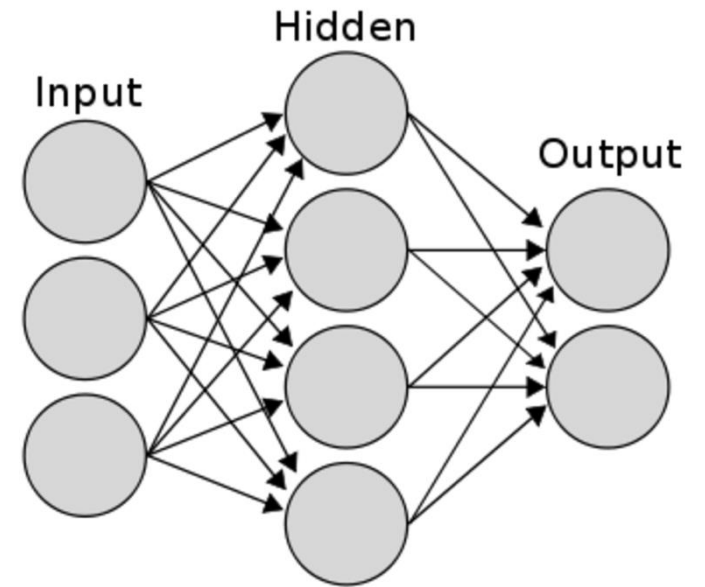
Apr 13, 2026

Recap: From kernels to neural networks



Recap: Two-layer neural networks

- Neural network: $S(x) = w_2^T (W_1 x + \mathbf{b}_1) + b_2$
 - Still a linear model at the end of the day, so let's add a nonlinearity σ !
- Two-layer MLP: $S(x) = w_2^T \sigma(W_1 x + \mathbf{b}_1) + b_2$
 - Linear model w.r.t. to a learnable feature map



Recap: Learning $\approx$ Configuring the learnable function so it behaves as instructed.

- Speech Recognition

$$f(\text{audio waveform}) = \text{"Hello!"}$$

- Handwritten Recognition

$$f(\text{handwritten digit 2}) = \text{"2"}$$

- Weather forecast

$$f(\text{☀️ Thursday}) = \text{"☁️🌧️ Saturday"}$$

- Play video games

$$f(\text{Sokoban game state}) = \text{"move left"}$$

Unsupervised learning

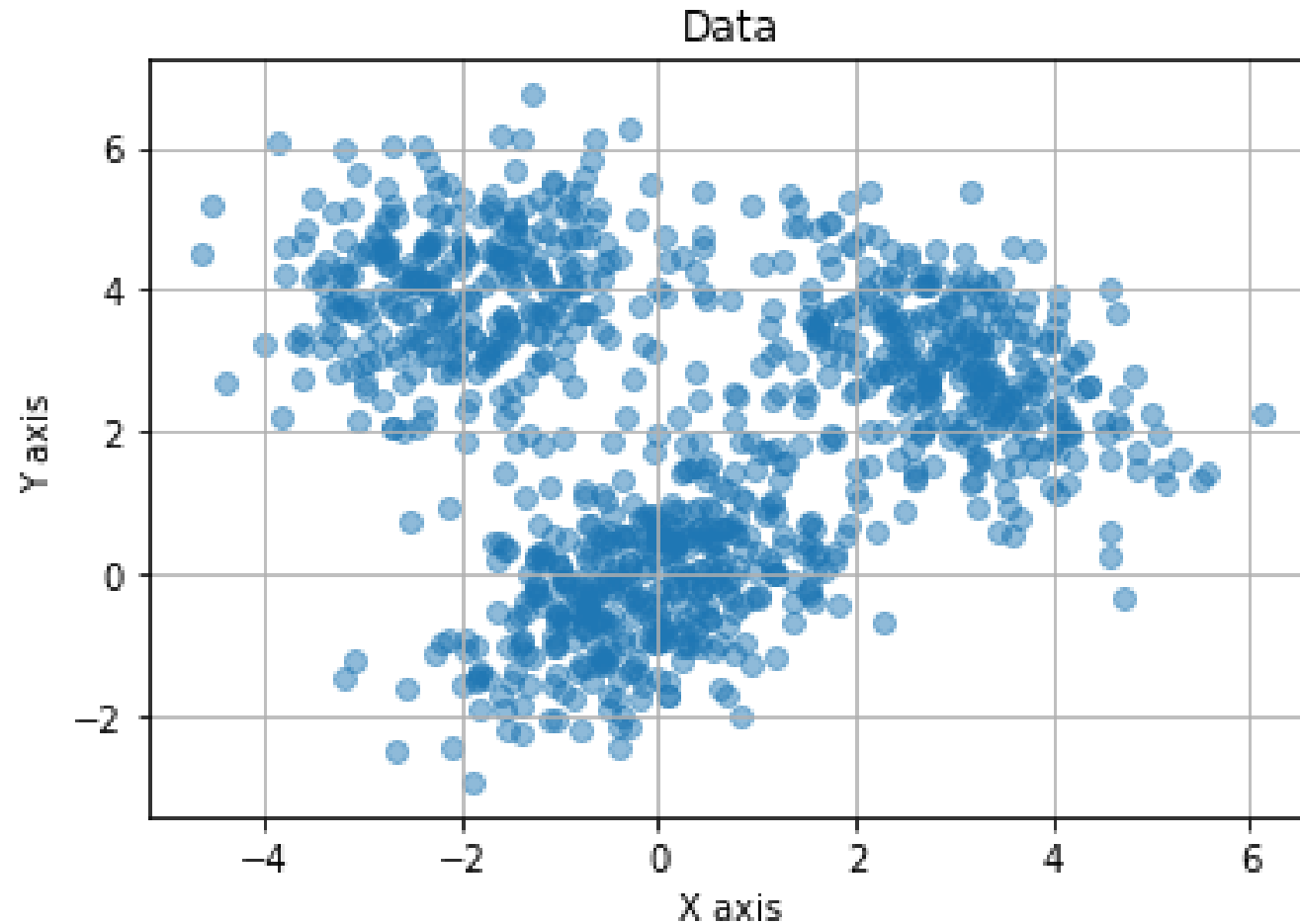
- Input space: $\mathcal{X}$ Images, videos, text, graphs, proteins, programs, etc...
- Output space: None.
- Hypothesis space: $\mathcal{H}$
- Each hypothesis h is a particular way to summarize the data
- Loss function $\ell : \mathcal{H} \times \mathcal{X} \rightarrow \mathbb{R}$
- Goal:
 - Discover data structure
 - Often achieved by minimizing the loss

Goal of unsupervised learning is to **learn data structures** without labels

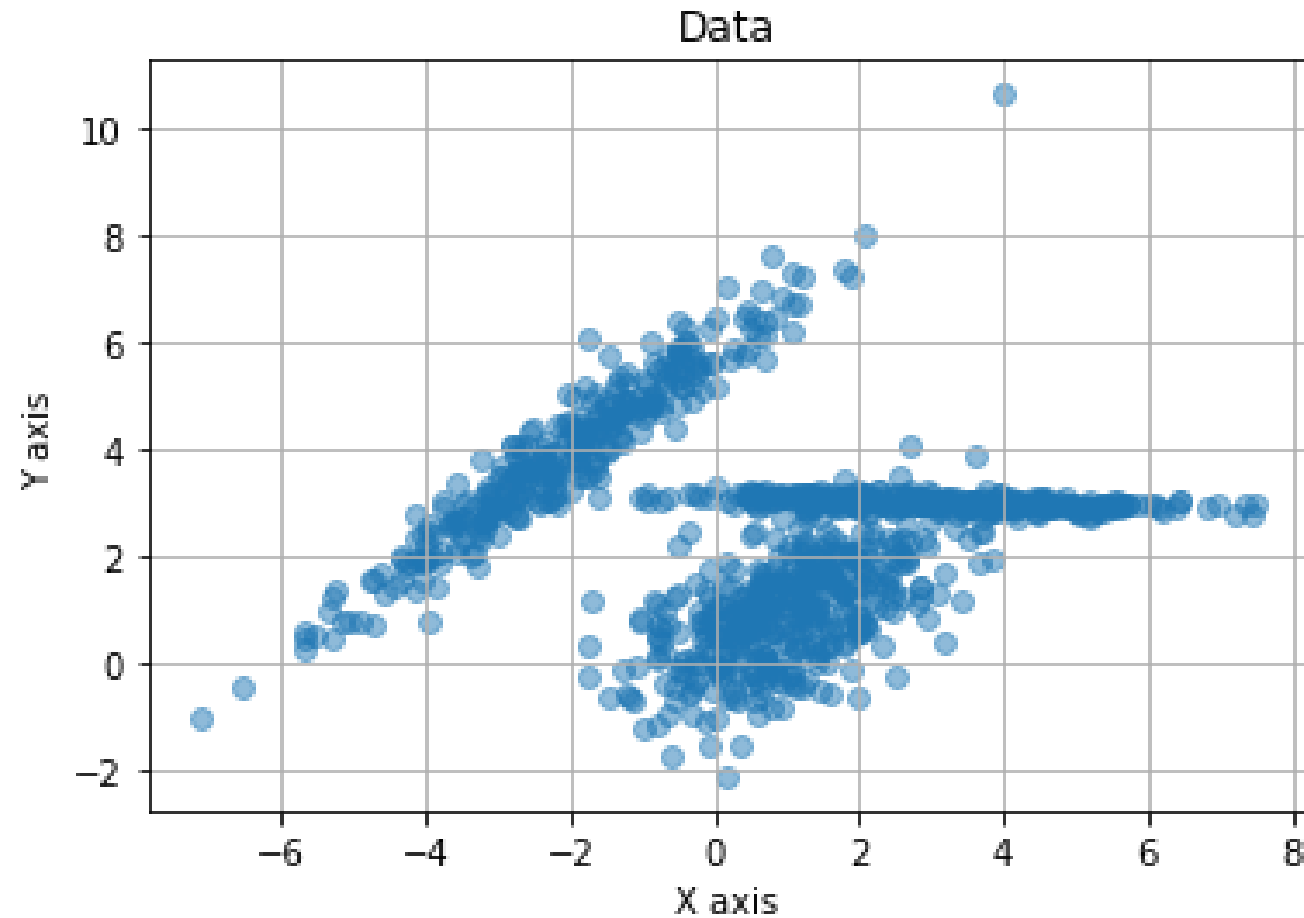


- Discussion: What kind of structures can you see?

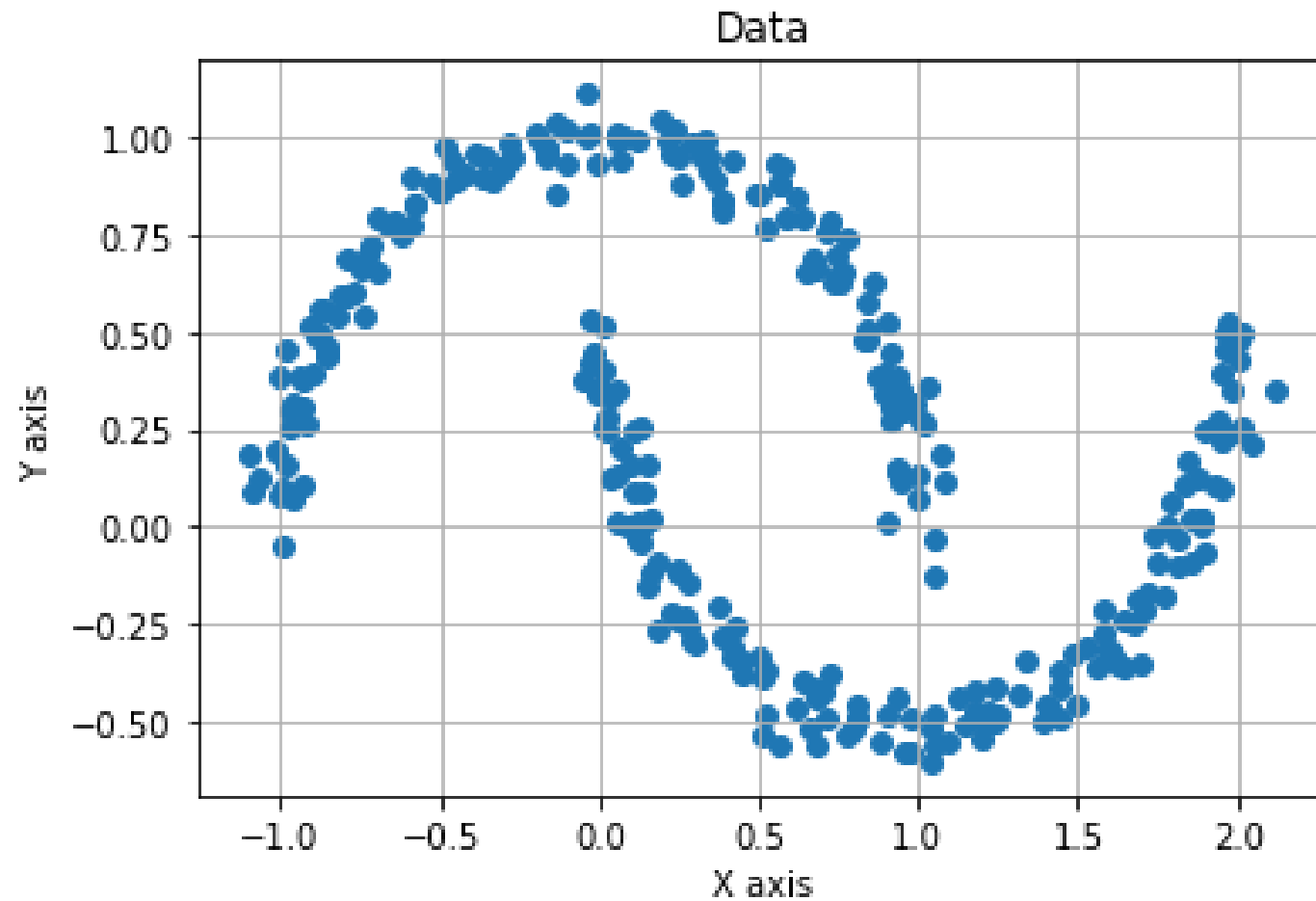
What kind of structures can you see?



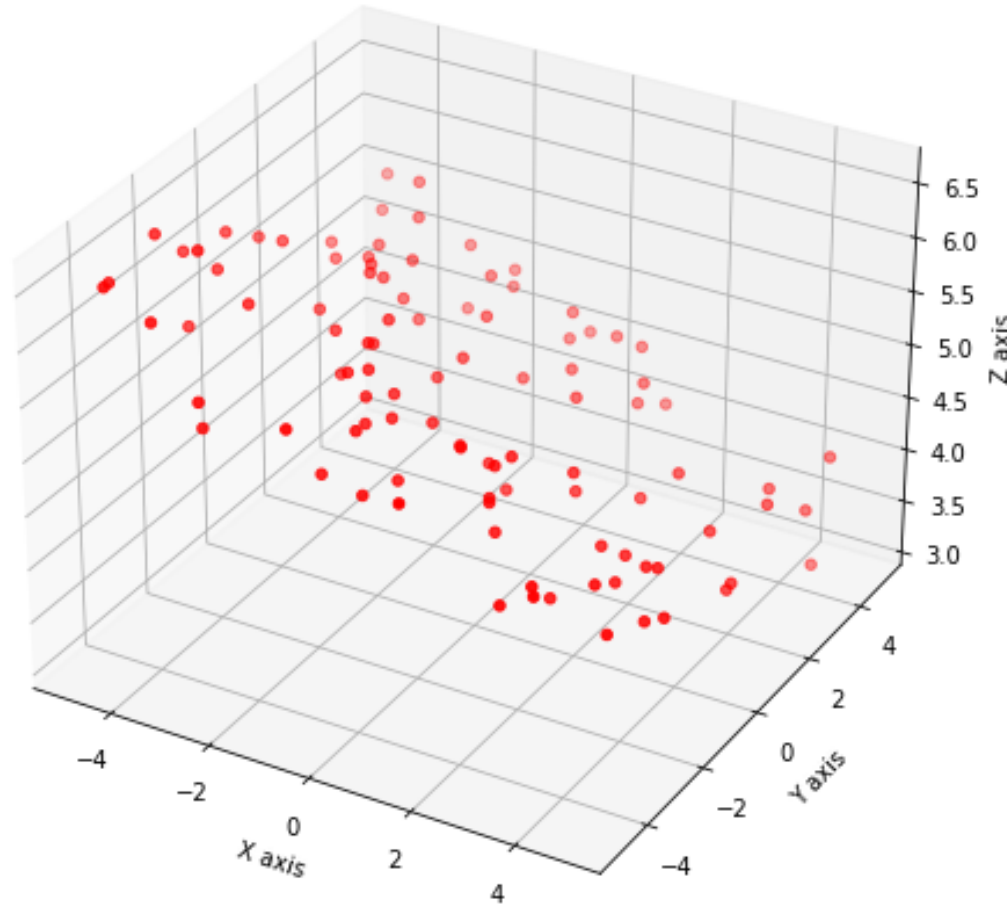
What kind of structures can you see?



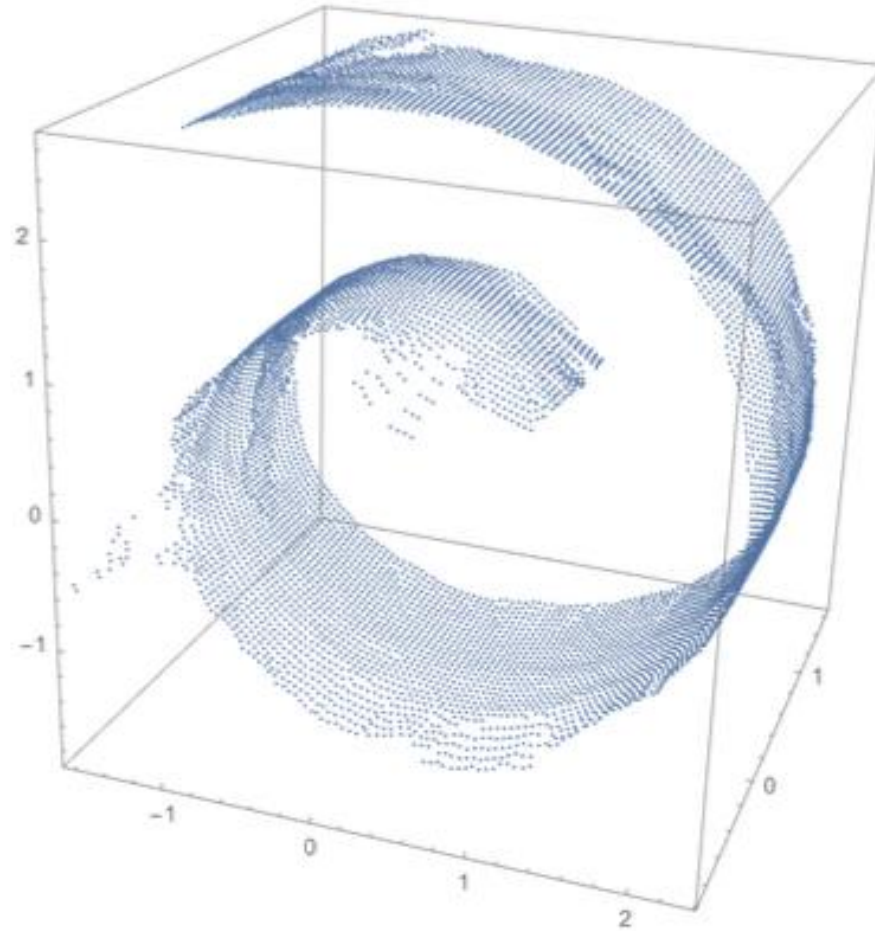
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What kind of structures can you see?



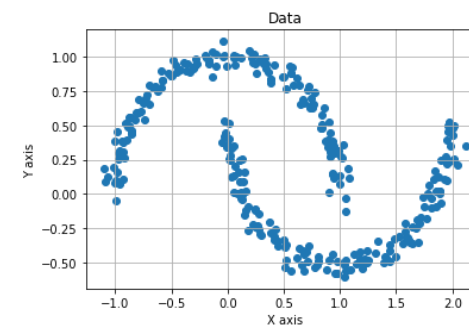
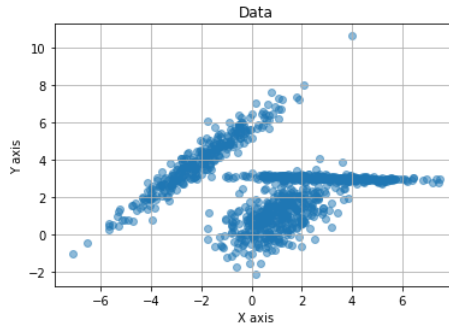
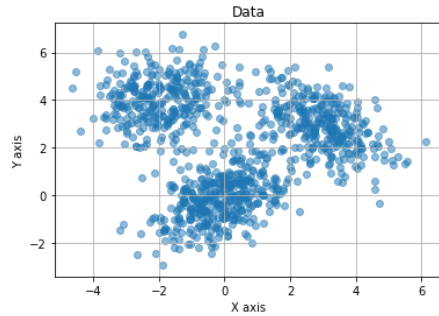
What kind of structures can you see?



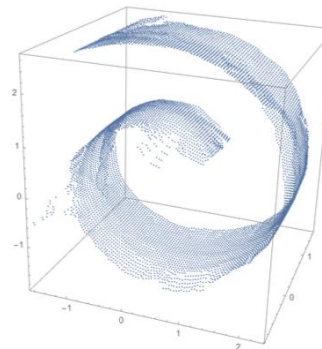
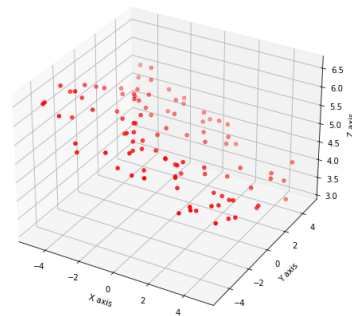
Two broad categories of unsupervised learning

(1) Clustering (2) Dimension reduction

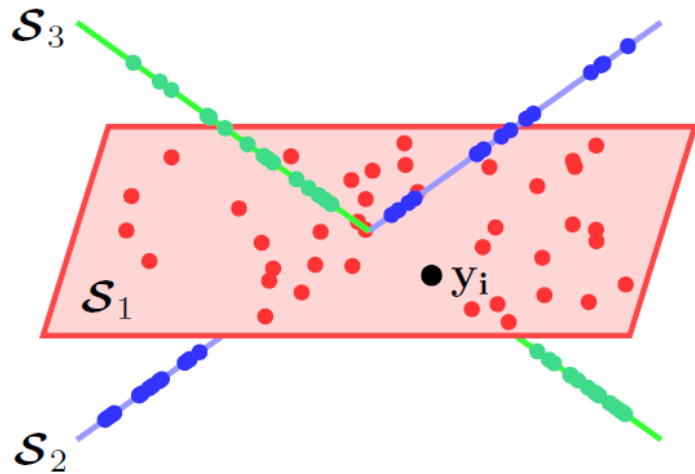
- Clustering:
 - finding a partition of the data that makes sense.



- Dimension reduction:
 - identifying a more compact representation (low-dimension) of data

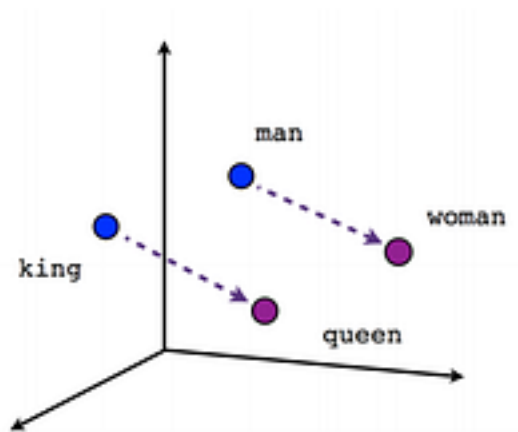


Application: Motion segmentation and subspace clustering

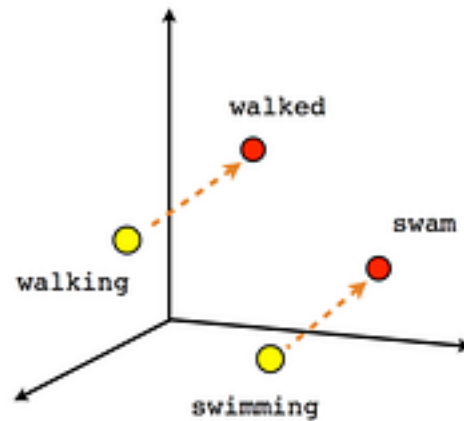


Applications: learn useful vector space representation of language

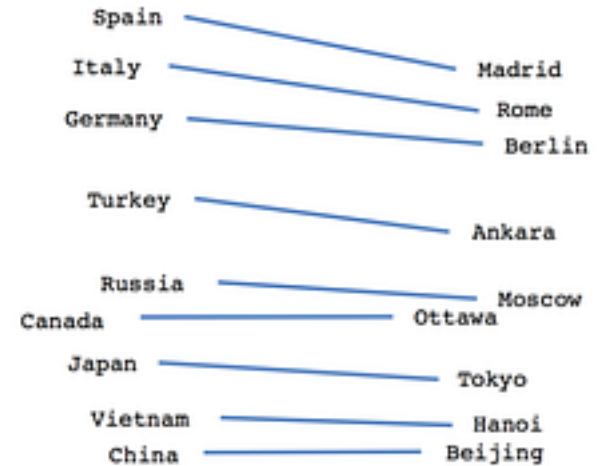
- So you can do algebra on them..



Male-Female



Verb tense



Country-Capital

Application: Image / video compression



100 dpi low JPEG compression



File size:
248K



100 dpi medium JPEG compression



File size:
49K

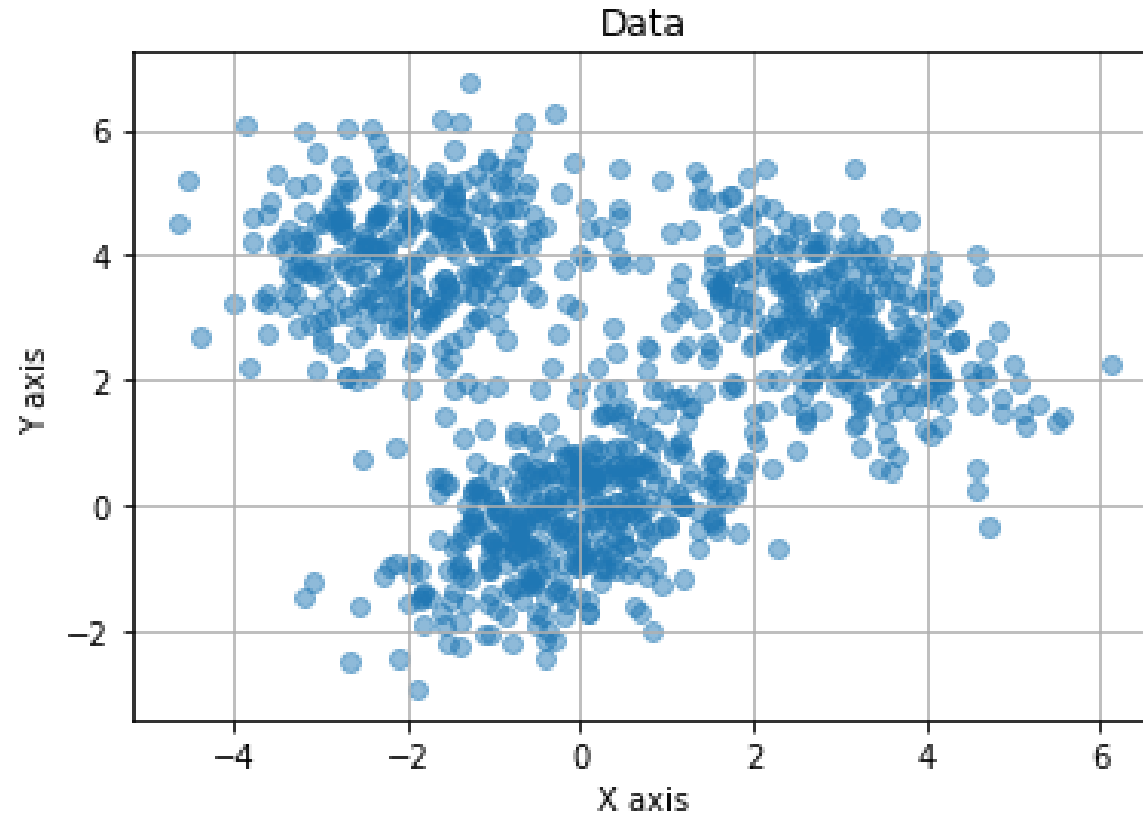


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22K

How do you learn the structure you see?



- Come up with a loss function to minimize?
- Come up a probabilistic model that generates the data?

The problem of k-means clustering

$$\arg \min_{\mathbf{S}} \sum_{i=1}^k \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2$$

- Where $\mathbf{S} = \{S_1, S_2, \dots, S_k\}$ is a partition of the dataset $(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$,
- And $\boldsymbol{\mu}_i = \frac{1}{|S_i|} \sum_{\mathbf{x} \in S_i} \mathbf{x}$, is called a cluster center (centroid) of S_i

The above optimization problem is equivalent to the following loss minimization

$$\min_{\mu_1, \dots, \mu_k \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n \min_{j \in [k]} \|x_i - \mu_j\|^2$$

- Once we find the **centroids**, finding the **partition of the data** is easy.
- If we have the **partition**, finding the corresponding **centroids** is also easy.
- **Idea:** Alternating minimizing the **centroids** and **cluster assignments**.

K-means clustering with Lloyd's algorithm

K is a hyperparameter

Algorithm $KMeans(D, K)$ – K -means clustering using Euclidean distance Dis_2

Input : data $D \subseteq \mathbb{R}^d$; number of clusters $K \in \mathbb{N}$.

Output : K cluster means $\mu_1, \dots, \mu_K \in \mathbb{R}^d$.

randomly initialise K vectors $\mu_1, \dots, \mu_K \in \mathbb{R}^d$;

repeat

 assign each $\mathbf{x} \in D$ to $\arg\min_j Dis_2(\mathbf{x}, \mu_j)$; *← 1-Nearest neighbor assignment*

for $j = 1$ to K **do**

$D_j \leftarrow \{\mathbf{x} \in D \mid \mathbf{x} \text{ assigned to cluster } j\}$; *← Partition defined by assignment*

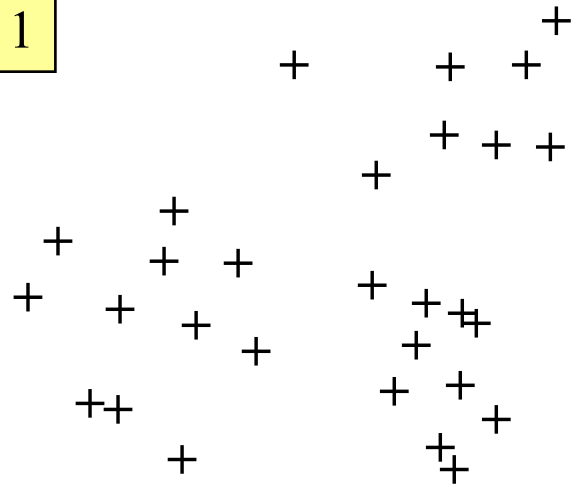
$\mu_j = \frac{1}{|D_j|} \sum_{\mathbf{x} \in D_j} \mathbf{x}$; *← Re-compute the cluster mean*

end

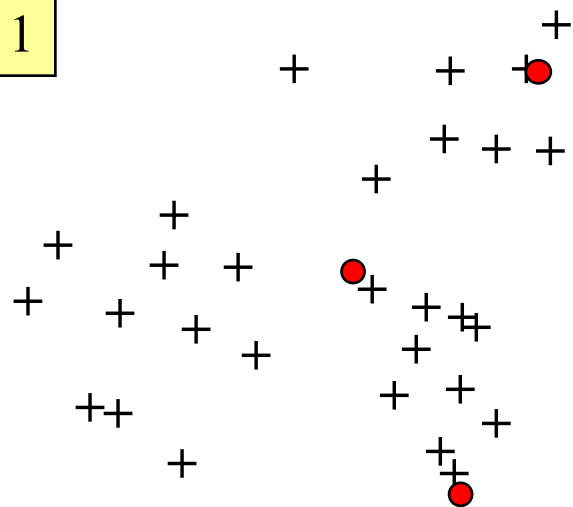
until no change in $\mu_1, \dots, \mu_K$;

return $\mu_1, \dots, \mu_K$;

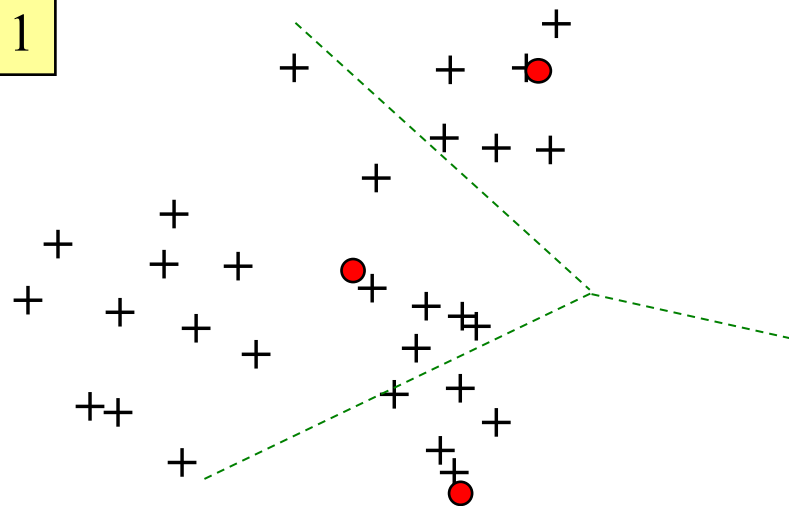
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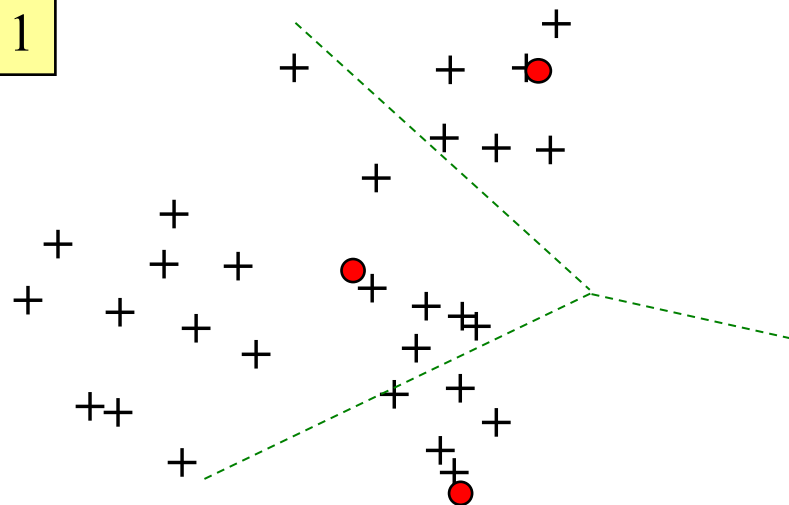
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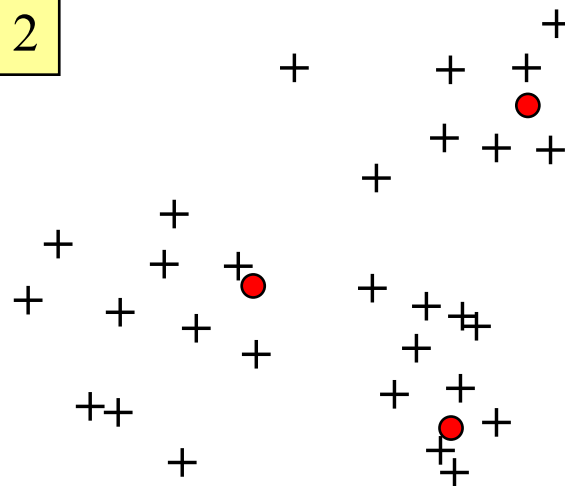
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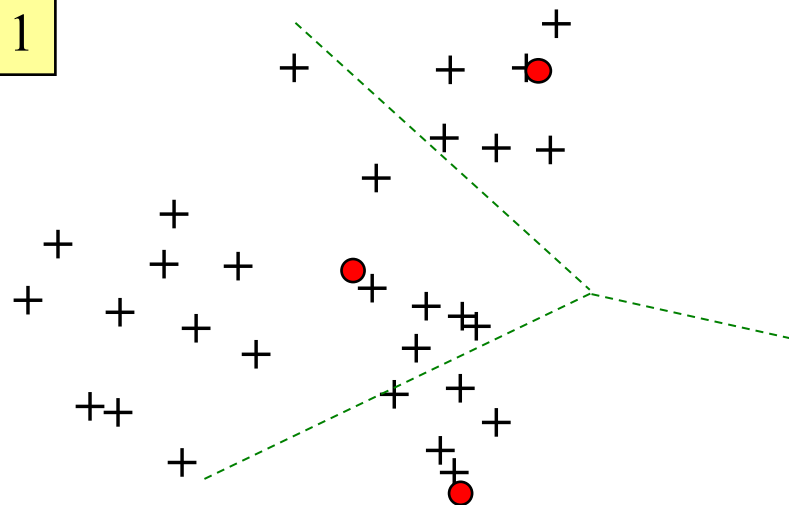
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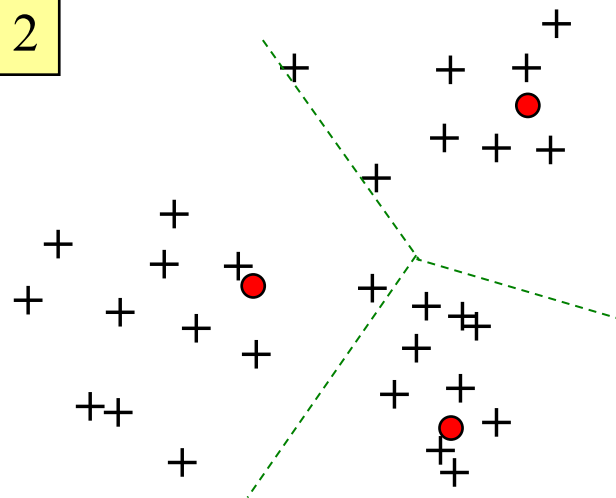
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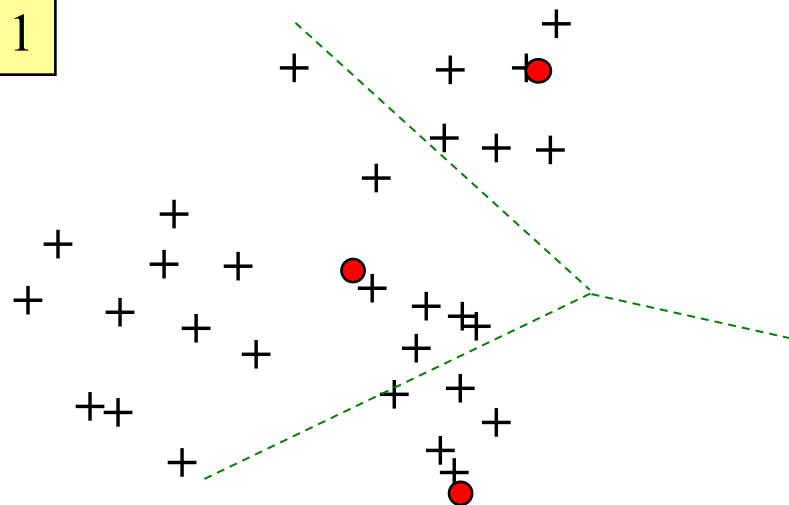
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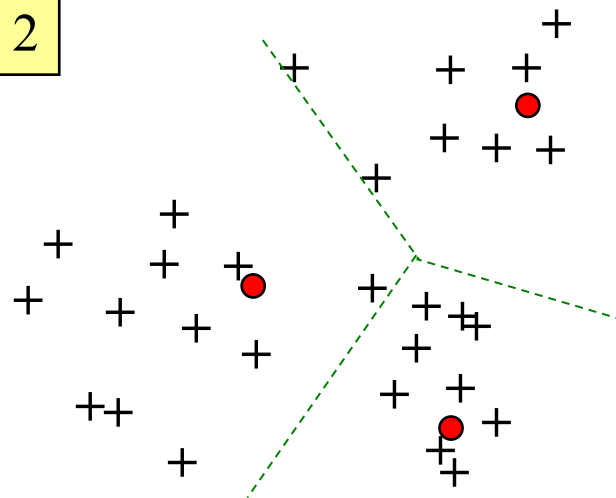
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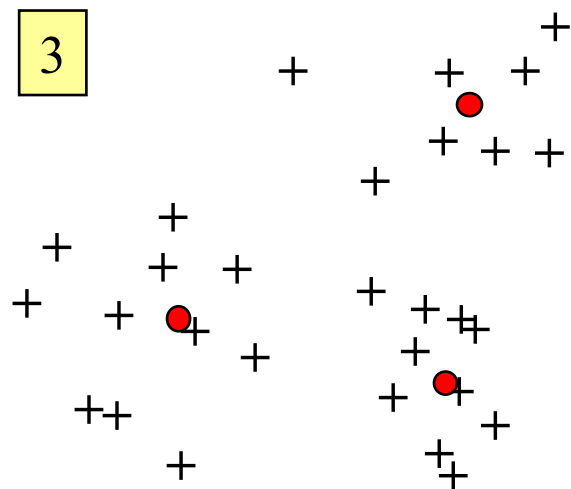
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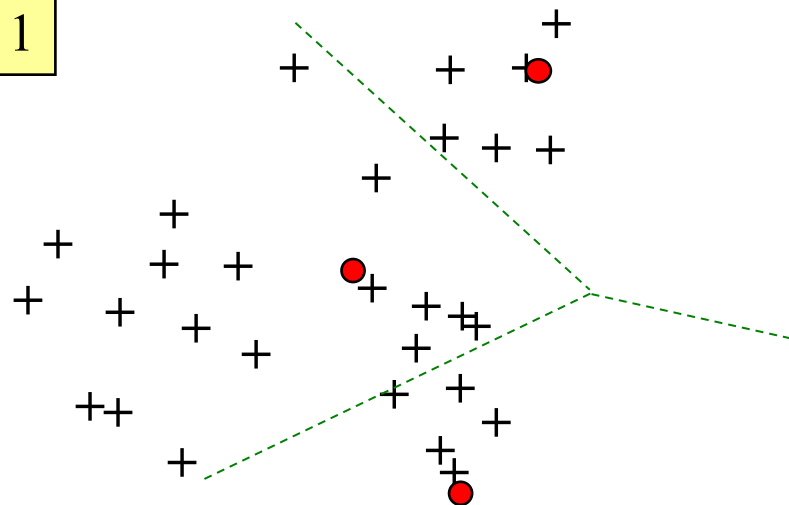
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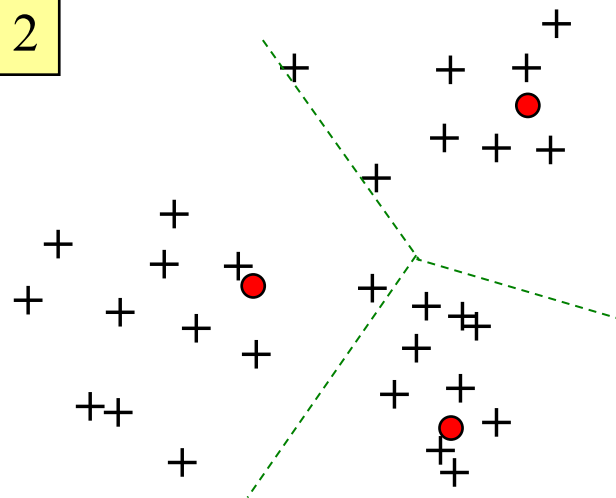
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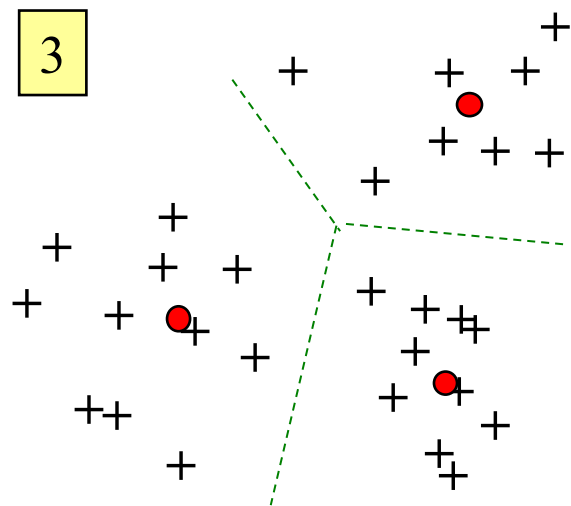
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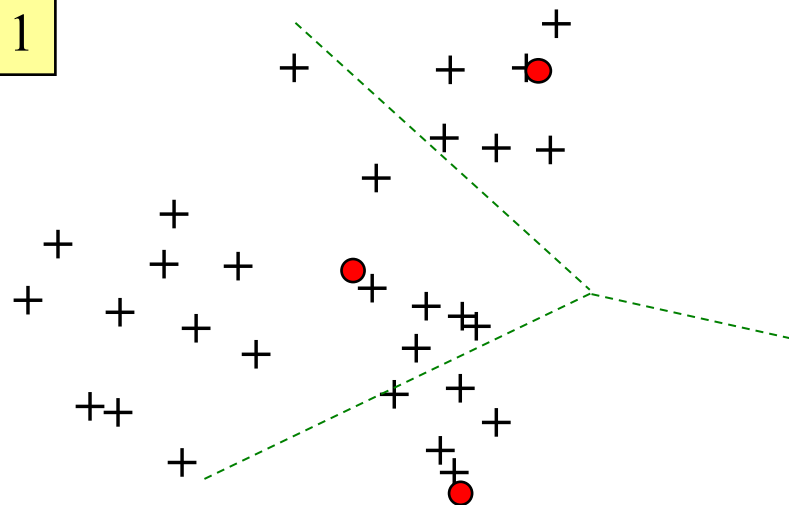
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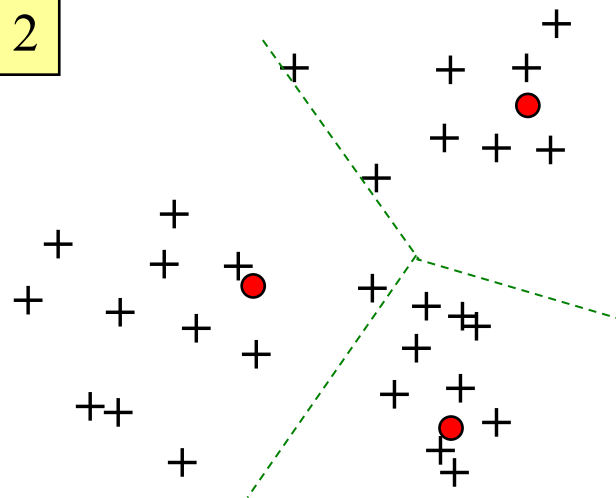
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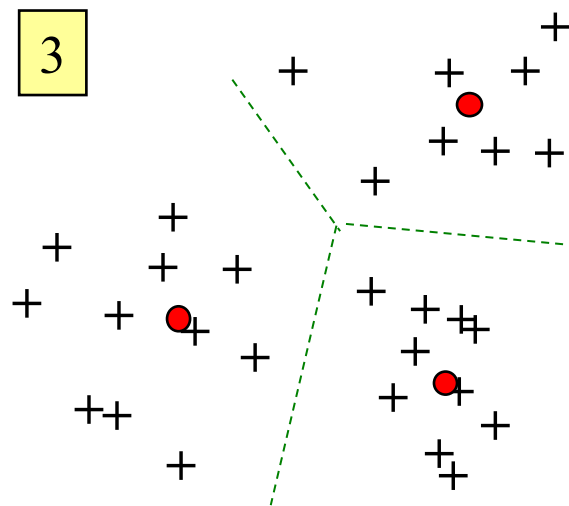
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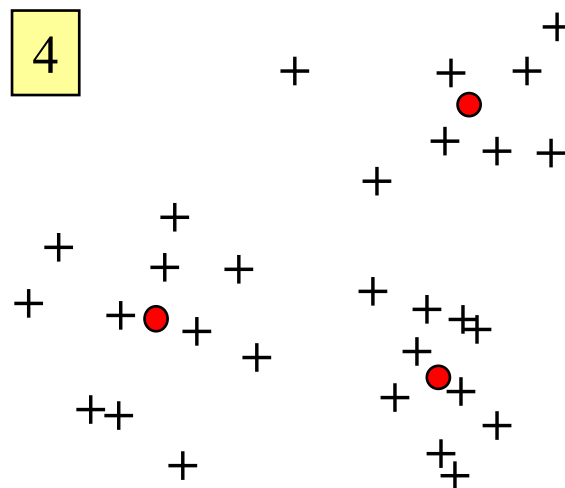
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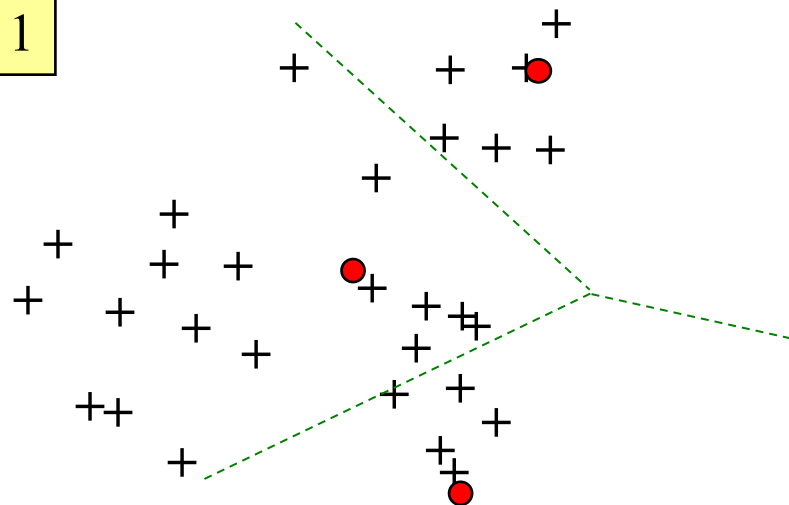
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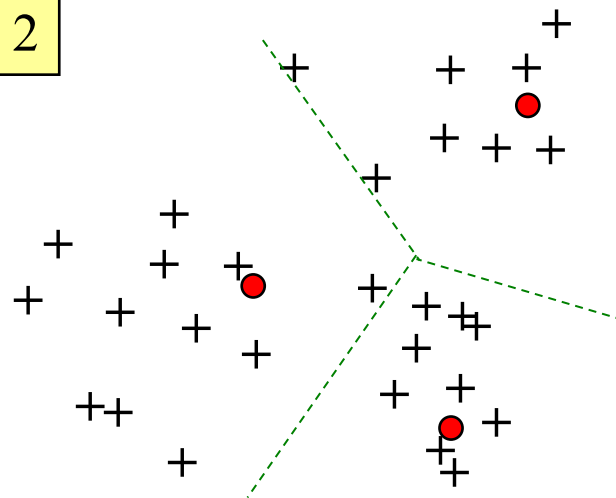
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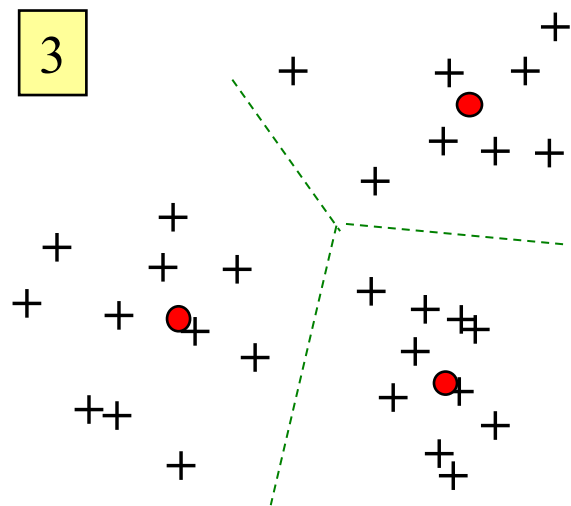
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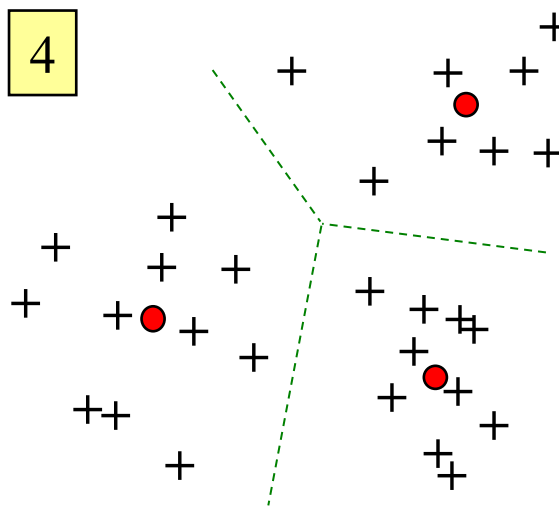
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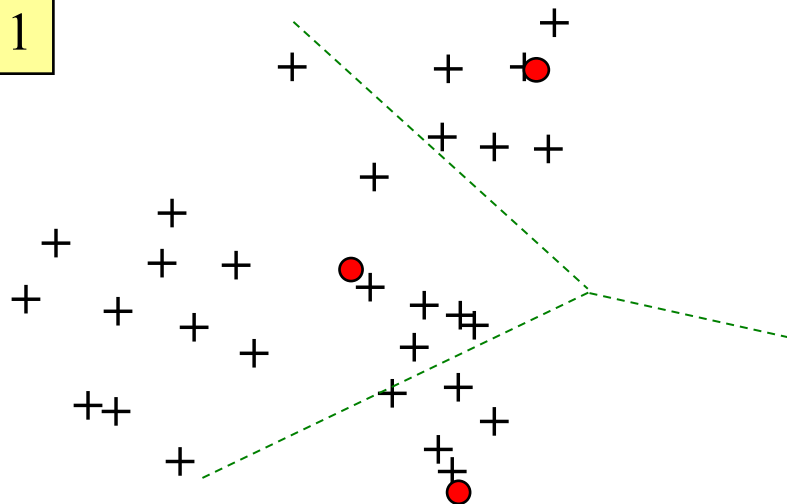
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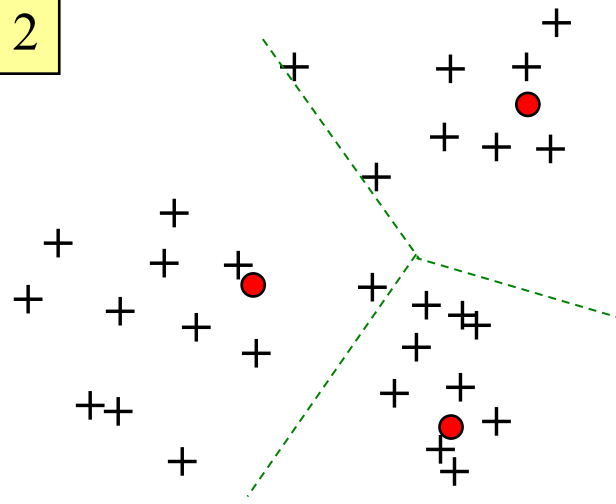
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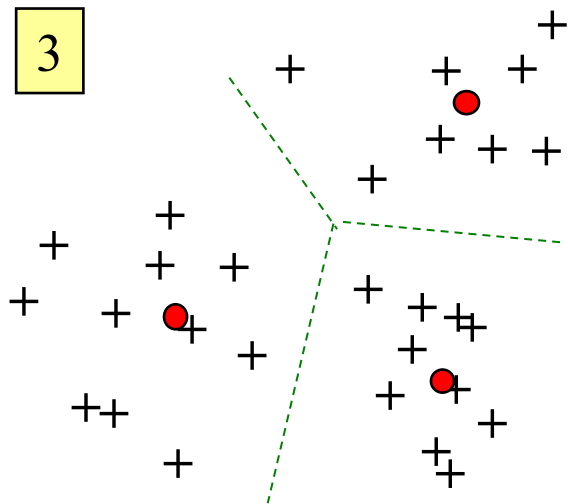
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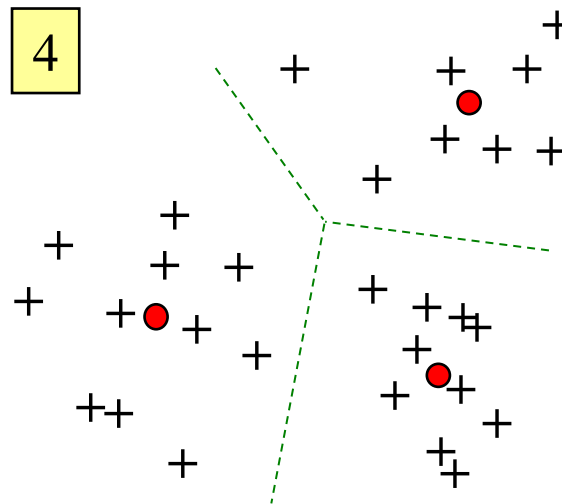
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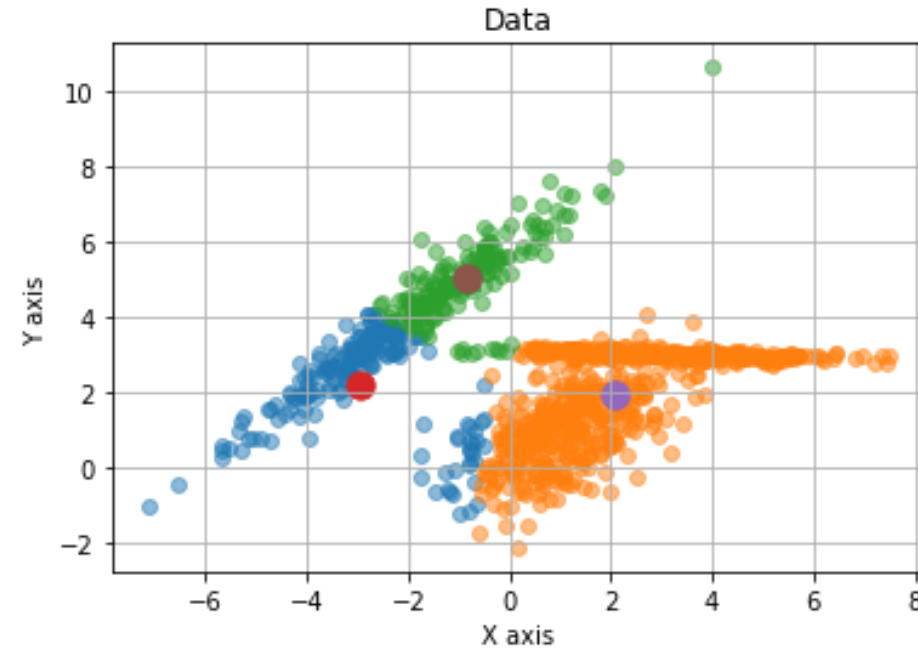
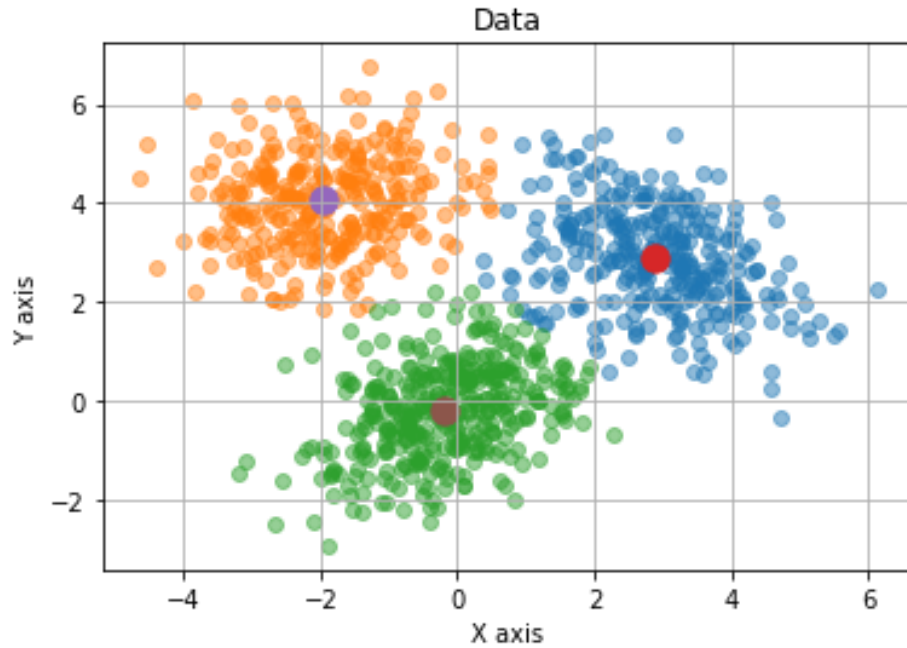


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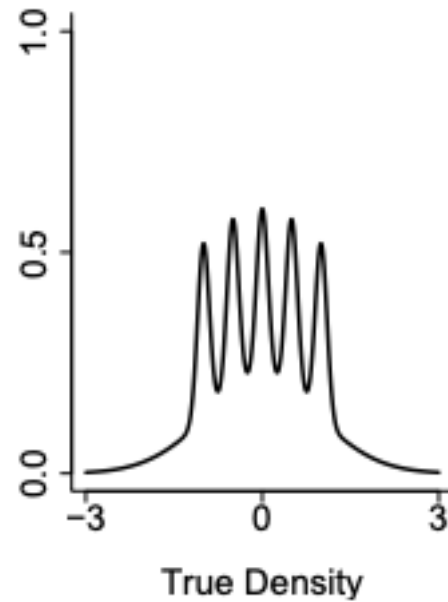
No change – finished!

K-means on our previous examples

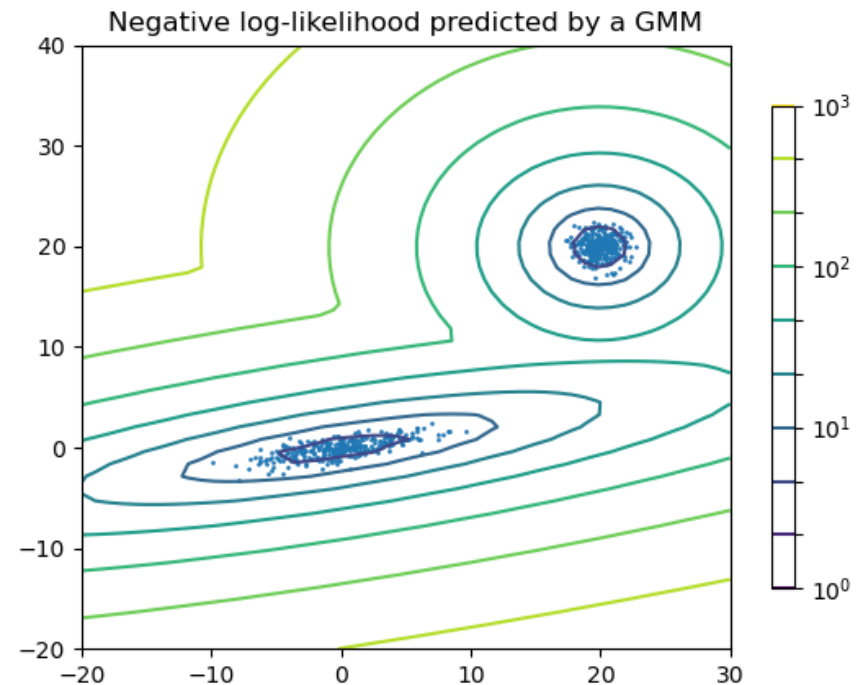


Gaussian mixture models

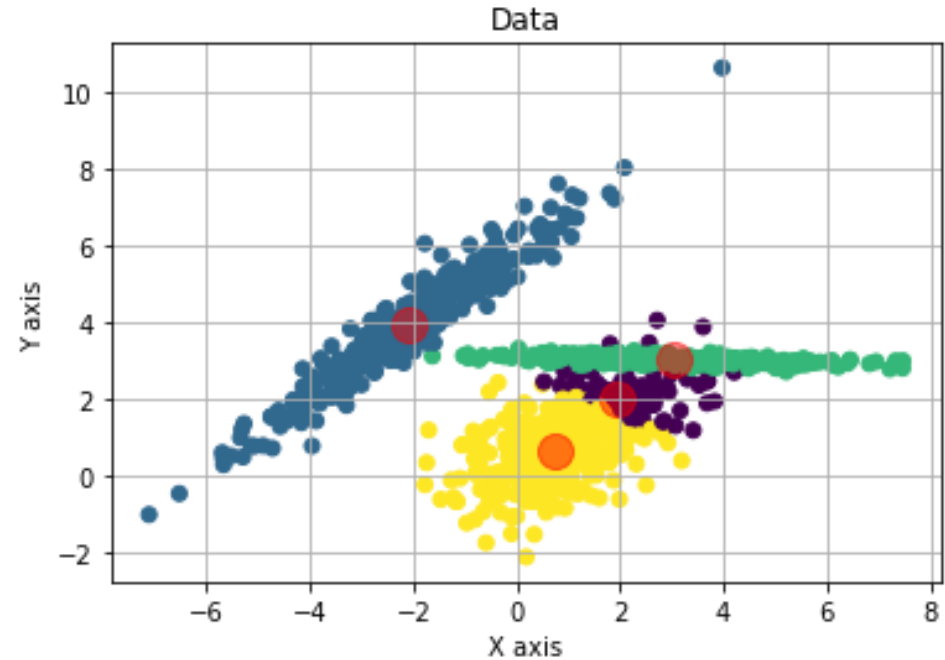
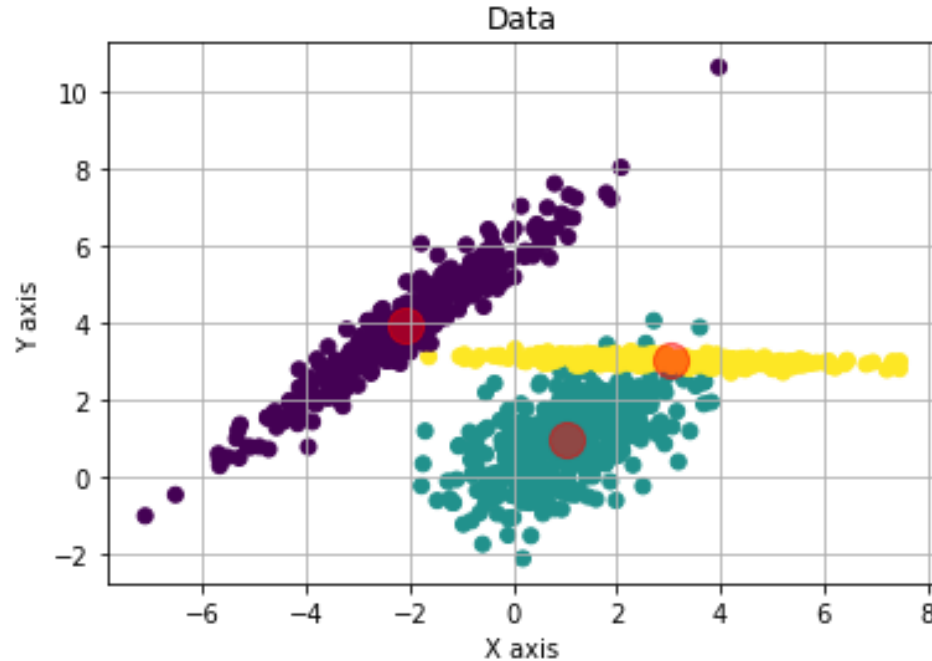
- Assume the data is generated from a mixture of Gaussian distribution



- Data generating process



Fitting mixture of Gaussian model



- Assign **soft** labels to each data point.
- Algorithms for fitting Gaussian mixture models?
 - **Expectation-Maximization** (not covered in this course... but also alternating making updates)

Summary: Unsupervised Learning

- K-means algorithm
 - Assign hard labels to data points
 - How does it work?
 - Alternating makes updates
 - Which distance function to use?
 - How many cluster centers (centroids) to choose?
 - How to initialize the centroids?
- Gaussian mixture models
 - Assign soft labels to data points
 - A probabilistic model for clustering

Announcement

- Teaching evaluation
 - Starting next Monday!
 - End on May 6
 - **All** students get 2 participation points if our response rate goes **above 60%!**
- Homework 4 will be released this Wednesday.
 - Due in 2 weeks.